TEAM RUDRAKSHA

ABHIYAN: Designing The Future

SMARTPRICE AI

NEPSE Stock Market Prediction System

A web app that uses machine learning to predict NEPSE stock prices and deliver Buy/Sell guidance.

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01 · PROBLEM

Problem Statement

Emotion-Driven Decisions

Many Nepalese investors trade based on emotion, tips, or rumours rather than data leading to poor outcomes.

No Localised Tools

Limited access to simple, localised prediction tools designed specifically for NEPSE-listed stocks.

Need for Transparency

Retail investors and students need a lightweight, explainable system to make data-driven decisions.

Objectives

01

Predict NEPSE Prices

Build an ML model to forecast next-day Close, High and Low prices for NEPSE-listed stocks.

02

Generate Buy / Sell Signals

Translate price predictions into simple, actionable guidance for retail investors and students.

03

Accessible Web Platform

Deploy a lightweight Streamlit app that anyone can use without financial or technical expertise.

04

Explainable & Transparent AI

Use interpretable Linear Regression so users understand how predictions are made.

Market Context & Target Users

NEPSE Market Snapshot

6.5M+

Demat account holders in Nepal

~254

Companies listed on NEPSE

NPR 90B+

Average daily turnover (2023–24)

Growing

Retail investor participation YoY

Who Is This For?

👤 Retail Investors

First-time or small-scale NEPSE investors lacking analytical tools.

🎓 Students & Researchers

Finance and CS students learning AI applied to real markets.

📊 Market Analysts

Professionals wanting a quick data-driven second opinion.

💼 Small Brokerages

Brokerage firms needing lightweight prediction tools.

Existing Solutions & Limitations

Manual Analysis

- Time-consuming
- Highly subjective
- No data consistency

Spreadsheet Models

- Requires expertise
- No automation
- Error-prone entry

Enterprise Platforms

- Very expensive
- Not localised for NEPSE
- Hard to use

Why SMARTPRICE AI Fills the Gap

- ✅ Free & open — no cost barrier for students or retail investors
- ✅ NEPSE-specific — trained exclusively on local market data
- ✅ Interpretable — Linear Regression lets users understand each prediction
- ✅ Simple UI — Streamlit app requires zero technical knowledge to operate

Our Solution

SMARTPRICE AI is a web-based application that uses 5 years of historical NEPSE data and a trained Linear Regression model to predict the next trading day's prices (Close, High, Low) and provide a clear Buy/Sell signal.

Model Core

Linear Regression built with scikit-learn — simple, fast, and interpretable for students.

- 5 Years of NEPSE Data
- 1,149 Rows / Stock

Web App

Streamlit UI for easy interaction. Enter today's inputs, get instant predictions & guidance.

- Buy / Sell Signal
- Dark Theme UI

How the System Works

01

Collect Data

5 years of NEPSE historical data for 3 stocks

02

Compute Features

MA-5, MA-10, Momentum, Daily Range, Prev High/Low

03

Train Model

Linear Regression - 80/20 train/test split

04

Predict & Signal

Output Close, High, Low + Buy/Sell recommendation

System Architecture

User
Web Browser

▶

Frontend
Streamlit UI
HTML / CSS

▶

Backend
Python Logic
Feature Engine

▶

AI Model
Linear Regression
scikit-learn

▶

Output
Predicted Prices
Buy/Sell

Tech Stack Frontend: Streamlit + HTML/CSS | Backend: Python · Pandas · NumPy | Model: scikit-learn | Persistence: Joblib

01 User Input — User enters today's Close, High, Low via Streamlit sidebar selectors.

02 Feature Computation — Backend calculates MA-5, MA-10, Momentum, Daily Range, Prev High/Low.

03 Model Inference — Joblib loads pre-trained Linear Regression model and runs prediction instantly.

04 Result Display — Streamlit renders Predicted Close/High/Low and Buy/Sell recommendation.

Dataset & Feature Engineering

NABIL	NTC	CIT
Nabil Bank	Nepal Doorsanchar	Citizen Inv. Trust
1,148 rows	1,149 rows	1,149 rows
R^2 0.91	R^2 0.87	R^2 0.94

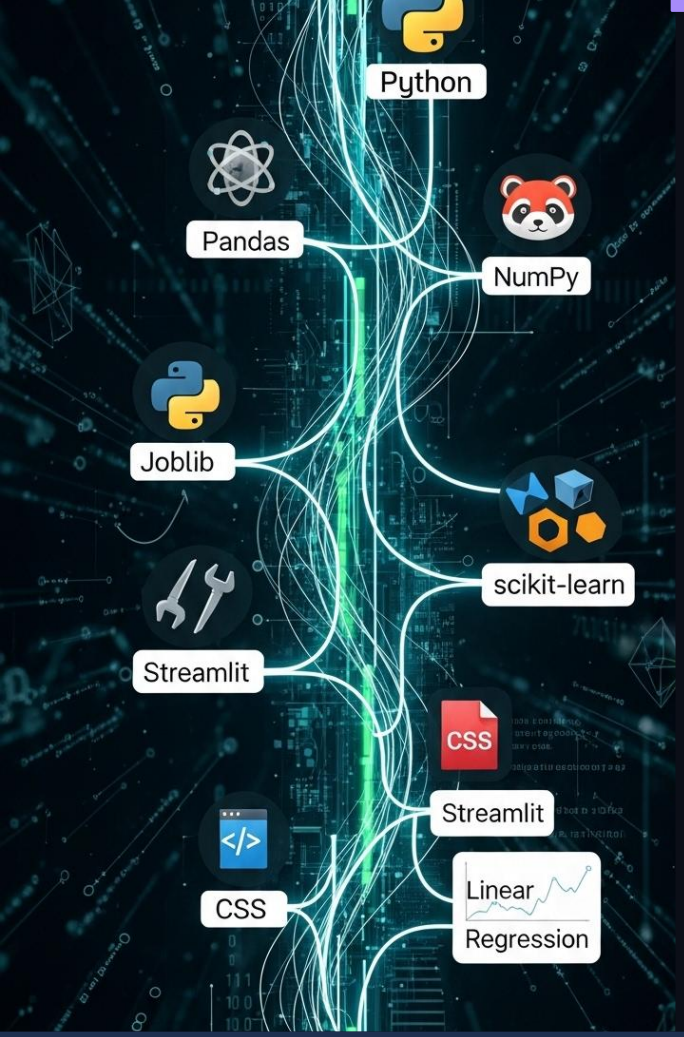
Engineered Features:

- PREV_CLOSE
- MA_5
- MA_10
- DAILY_RANGE
- MOMENTUM
- PREV_HIGH
- PREV_LOW

Data Source: https://nepsealpha.com/nepse-data

5 YEARS

Data for training



Technologies Used

Python	Core programming language
Pandas & NumPy	Data cleaning and numerical feature computation
Scikit-learn	Linear Regression model for price prediction
Joblib	Save and reload trained models efficiently
Streamlit	Quick, interactive web interface for live demos
HTML/CSS	Custom UI styling for a modern AI look

Features of SMARTPRICE AI

Price Predictions

Predicted Close, High and Low for the next trading day.

Buy/Sell Signal

Simple action recommendation based on predicted movement.

Feature Set

MA (5,10), Momentum, Daily Range, Previous Highs/Lows.

Historical Coverage

Uses 5 years of NEPSE data to capture seasonality and longer-term trends.

Model Performance Results

Algorithm

Linear Regression

Train/Test: 80% / 20% · shuffle = False

0.94
Best R^2

99%+
Accuracy

3.12
Min MAE

3
Stocks

Stock	MAE	RMSE	R^2 Score	Accuracy
Nabil Bank	3.12	5.06	**0.9104**	99.38%
Nepal Doorsanchar	5.48	8.15	**0.8724**	99.37%
Citizen Inv. Trust	11.50	18.98	**0.9395**	99.39%

Business Impact

📉 Reduced Emotional Trading

Data-backed signals help investors avoid costly impulsive decisions.

🏦 Democratised Market Access

Free, localised tool levels the playing field for Nepal's retail investors.

⚡ Faster Decision Making

Instant next-day predictions eliminate hours of manual chart analysis.

📚 Education & Awareness

Transparent Linear Regression helps students learn real-world AI.

📈 Improved Forecast Accuracy

99%+ prediction accuracy across 3 stocks vs. gut-feel analysis.

🌐 Scalable Foundation

Architecture supports LSTM models, more stocks, and live NEPSE API feeds.

Predicted Low: 748.50

Live Demo Explanation

1. Select stock (e.g., Nabil Bank) and input today's close and optional technical indicators.
2. Click Predict Streamlit sends data to the loaded Joblib model instantly.
3. Model returns Predicted Close, High, Low and a clear Buy/Sell signal.
4. Results shown with confidence range and a brief explanation for learners.

Designed to be fast, interpretable and easy to use during the live presentation.

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SMARTPRICE AI

Live Prediction Test Report

9	89%	Rs. 8.43
Tests Run	Signal Accuracy	Avg Close Error

Stocks Tested: NABIL Bank · Nepal Doorsanchar (NTC) · Citizen Investment Trust (CIT)

NABIL

NABIL Bank — Test Results

Predict Date	Input Date	Close Err	High Err	Low Err	Signal
2026-02-01	2026-01-29	1.68	3.30	2.70	BUY ✅
2026-02-09	2026-02-08	2.43	2.88	3.98	SELL ✅
2026-03-03	2026-03-01	4.08	4.26	1.64	BUY ✅
AVERAGE		**2.73**	**3.48**	**2.77**	**3/3** ✅

✅ All 3 signals correct — most consistent stock. Avg close error Rs. 2.73

NTC

Nepal Doorsanchar (NTC) — Test Results

Predict Date	Input Date	Close Err	High Err	Low Err	Signal
2026-02-01	2026-01-29	3.19	2.11	2.29	BUY ❌ (flat)
2026-03-01	2026-02-26	10.89	8.29	0.71	BUY ✅
2026-03-03	2026-03-01	7.03	0.36	5.14	BUY ✅
AVERAGE		**9.70**	**3.59**	**2.71**	**2/3 ✅**

⚠️ 1 flat-day miss (Rs. 0.81 move). Big-move signals 100% correct.

CIT

Citizen Investment Trust (CIT) — Test Results

Predict Date	Input Date	Close Err	High Err	Low Err	Signal
2026-02-16	2026-02-12	13.19	0.68	14.32	SELL ✅
2026-03-01	2026-02-26	14.49	15.33	24.17	BUY ✅
2026-03-03	2026-03-01	10.92	11.24	5.24	BUY ✅
AVERAGE		**12.87**	**15.75**	**14.58**	**3/3 ✅**

✅ All signals correct. Higher errors on breakout days (Rs. 10+ jumps).

SUMMARY

Overall Test Results — All 9 Tests

Date	Stock	Close Err	High Err	Low Err	Signal
2026-02-01	NABIL	1.68	3.30	2.70	BUY ✅
2026-02-09	NABIL	2.43	2.88	3.98	SELL ✅
2026-03-03	NABIL	4.08	4.26	1.64	BUY ✅
2026-02-01	NTC	3.19	2.11	2.29	❌ flat
2026-03-01	NTC	10.89	8.29	0.71	BUY ✅
2026-03-03	NTC	7.03	0.36	5.14	BUY ✅
2026-02-16	CIT	13.19	0.68	14.32	SELL ✅
2026-03-01	CIT	14.49	15.33	24.17	BUY ✅
2026-03-03	CIT	10.92	11.24	5.24	BUY ✅
AVERAGE	**ALL**	**8.43**	**7.61**	**6.69**	**8/9 ✅**

Future Improvements

Deep Learning

Upgrade to LSTM/Transformer models for capturing temporal patterns.

Real-Time NEPSE API

Connect to live NEPSE feed for streaming predictions and alerts.

Expand Coverage

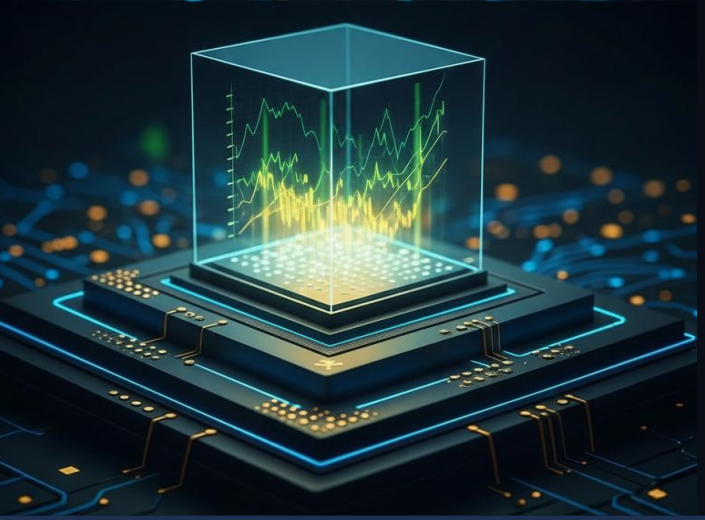
Add more NEPSE-listed companies and instrument-level models.

Visual Dashboards

Live charts, confidence intervals, and backtest result visualisations.

Ensemble Methods

Improve accuracy by combining multiple ML algorithms.



Conclusion

SMARTPRICE AI demonstrates a practical, student-built application of machine learning to Nepal's stock market. With 5 years of NEPSE data and an accessible web UI, it helps learners and retail investors make more informed decisions.

Website link: https://smartpriceproai.streamlit.app

3	5yr	0.94	99%+
Stocks	Data	Best R^2	Accuracy

Thank You!

Questions are welcome.

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